Disentangling Contextual Coherence and Prompt Echo in Early R-Lens Readouts

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CAMBRIA capstone — a blinded two-model independent evaluation and methodological extension of the R-Lens early-layer coherence claim

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Abstract

R-Lens is a variant of J-Lens that fits its layer-to-final-layer transport with LRP stop-gradients. Its authors report, qualitatively and without a released scorer, that R-Lens readouts at early layers are more coherent than J-Lens readouts. We evaluate that claim on two 27B models with a blinded, pre-specified and frozen rating panel of 200 three-arm cells, scored by two independently validated LLM autoraters.

Under a standard prompt-aware coherence rubric, R-Lens is rated more coherent than J-Lens on both models: +0.690, 95% CI $[0.520, 0.855]$ on Qwen3.5-27B and +1.010, $[0.785, 1.235]$ on Gemma-3-27B-it, on a 0–4 scale. R-Lens also surfaces significantly more prompt-overlapping content than J-Lens.

We therefore built a second, separately validated rubric that instructs judges to discard tokens copied from the prompt and score only what remains. Under it the R–J gap attenuates by 53% on Qwen, to +0.325 $[0.165, 0.480]$, and by 89% on Gemma, to +0.110 $[-0.045, 0.255]$, which no longer excludes zero. Both Jacobian-based lenses nevertheless remain clearly better than the logit lens on non-copied content.

The contribution is therefore not a simple confirmation. R-Lens's advantage under conventional contextual scoring is real and reproduces on both models, but it is substantially prompt-local, and on Gemma we cannot detect an advantage in novel non-copied content at all. We also show that restricting to cells with equal observed echo — the natural first response to this confound — gives a markedly more reassuring answer than direct measurement does, because it conditions on a variable the lens itself affects.

1 What this document does and does not claim

This is an independent evaluation of the coherence claim, and a methodological extension of it. It is *not* an exact replication: the source post released no scorer, so the rubric, panel and statistics here are ours, and the released lens artifacts we evaluate were fit with a lighter recipe than the one the post describes (§2).

It asks two questions. First, when a rater is shown three unlabelled early-layer readouts of the same prompt — one from the logit lens, one from J-Lens, one from R-Lens — does the R-Lens arm receive a higher contextual-coherence score? Second, does any advantage survive when the rater is forbidden from crediting content copied from the prompt?

It makes **no** claim about concept specificity, causal onset, ablation, necessity, sufficiency, answer smuggling, or whether any lens recovers a representation the model actually uses. Separate experiments in this project address some of those; by pre-specification their results are not pooled with, used to interpret, or reported alongside the coherence estimand, and they do not appear here.

Terminology. The protocol was written, frozen and hashed before any rating was collected, but it was not deposited with a third-party registry. We therefore say *pre-specified and frozen* throughout, never *preregistered*.

2 Artifacts and provenance

Both lenses are the authors' released artifacts, not refits, taken at a pinned commit. A preflight check refuses to run if either lens fails provenance:

- J-Lens carries `{"estimator": "standard"}`.
- R-Lens carries `{"estimator": "relp"}` with the LN rule, identity rule and half rule at $\beta = 0.5$.
- Both were fit with `n_prompts: 25`—the released light recipe, not the $n = 1000$ recipe described in the source post. Any difference between our effect size and theirs is confounded with this.
- The two Jacobian sets are verified to actually differ, so a null could not be produced by silently loading the same lens twice.

Preflight additionally verifies hook indexing (that recorded block outputs satisfy `activations`$[\ell]$ equals `hidden_states`$[\ell+1]$), target layer identity, and transport orientation. A failure at any of these is fail-closed: the run aborts rather than proceeding with a warning.

3 Method

3.1 Panel

Twenty prompts—four from each of the five official evaluation sets (association, multihop, multilingual, poetry, typo)—were selected by a hash of a fixed salt, the set name and a stable item identifier, so selection is reproducible and independent of any result. For the two sets carrying a target (multihop, multilingual) only items the model itself answers correctly were eligible, matching the source protocol's correctness filter. Prompts are shared between models: both models are read on the same 20 items.

Each prompt is read at five pre-specified relative depths $z = \ell/\ell^* \in \{0, 0.1, 0.2, 0.3, 0.4\}$, giving $20 \times 5 \times 2 = 200$ cells, each holding three arms. Readout position follows the source protocol: the final prompt token, except poetry, read at the newline ending line one.

3.2 Blinding

Arm order is permuted per *(cell, rater)* pair, not once per cell. A single fixed order would make position bias identical across judges and therefore invisible to any agreement statistic. The unblinding key was written outside the repository and outside the published result tree; the panel and the sample carry content hashes; and every outgoing judge payload was reconstructed after the fact and scanned for structural leakage of lens identity.

3.3 Rubric

Three dimensions per arm: **contextual coherence** (0–4, primary), **lexical integrity** (0–2) and **prompt echo** (0–2). Contextual coherence asks whether the readout tokens form a plausible continuation or elaboration of the prompt's content. Prompt echo measures how much of the readout is copied from the prompt — it is scored so that copying can be detected, not rewarded.

3.4 Judges

Two primary autoraters, `openai/gpt-5` and `deepseek/deepseek-chat-v3.1`, each independently validated on a 78-cell battery of control categories before being admitted. Validation gates include position-bias rate (tested with an exact binomial, so that a threshold finer than the achievable resolution at that sample size returns UNDERPOWERED rather than a false pass), a corruption-detection control, a duplicate-arm control, and a response-completeness gate that fails a judge returning empty content on more than 2% of cells. One candidate judge was rejected under exactly this last gate.

The primary score is the **mean of the two validated judges**. A third judge (`meta-llama/llama-3.1-70b-instruct`) was originally used to adjudicate the 63% of cells on which the primaries disagreed, producing a median-of-three score. That adjudicator did not clear validation, and the primary estimate was therefore demoted to the mean of two before any result was written up. Section 5.2 gives the gate it failed to resolve, and Section 5.3 reports the adjudicated estimate as a sensitivity analysis.

This change increases the reported effect (pooled $+0.778 \rightarrow +0.850$), and we state that explicitly because a reader is entitled to ask whether the rule was chosen for its result. It was not: the demotion follows from an instrument failing its admission gate, the gate was specified before the adjudicator was run, and the direction and significance of every contrast are unchanged either way. Both estimates appear in this document.

3.5 Statistics

The estimand is the equal-weight paired difference: mean over evaluation sets, then prompts within set, then depths within prompt, so that no set or prompt is over-weighted by having more cells. Intervals are percentile 95% from a 10,000 replicate stratified paired *prompt-cluster* bootstrap. p-values are paired prompt-cluster sign-flip permutation tests — never a bootstrap sign proportion. Depth and set breakdowns are Holm-corrected within model.

4 Results

4.1 Primary endpoint

R-Lens is rated more contextually coherent than J-Lens on both models (Table 1). Both intervals exclude zero in the positive direction, so the ordering holds on both models under this rubric.

Two features of this table matter as much as the headline.

J-Lens is not distinguishable from the logit lens on either model ($+0.045$, $p = 0.776$ on Qwen; $+0.430$, $p = 0.112$ on Gemma). On Qwen the two are almost exactly equal in mean score, 1.295 against 1.250 on a 0–4 scale. Under the primary rule the entire measured advantage over the logit-lens baseline is carried by the R-Lens variant, not by the Jacobian transport as such. Under the adjudicated rule the Qwen contrast is instead positive and significant, and Section 5.3 shows it is judge-unstable in sign; it should not be quoted as a single number in either direction.

On Gemma the bootstrap interval for J $-$ logit excludes zero ($[0.055, 0.790]$) while the permutation test does not reject ($p = 0.112$). The two procedures ask slightly different questions and

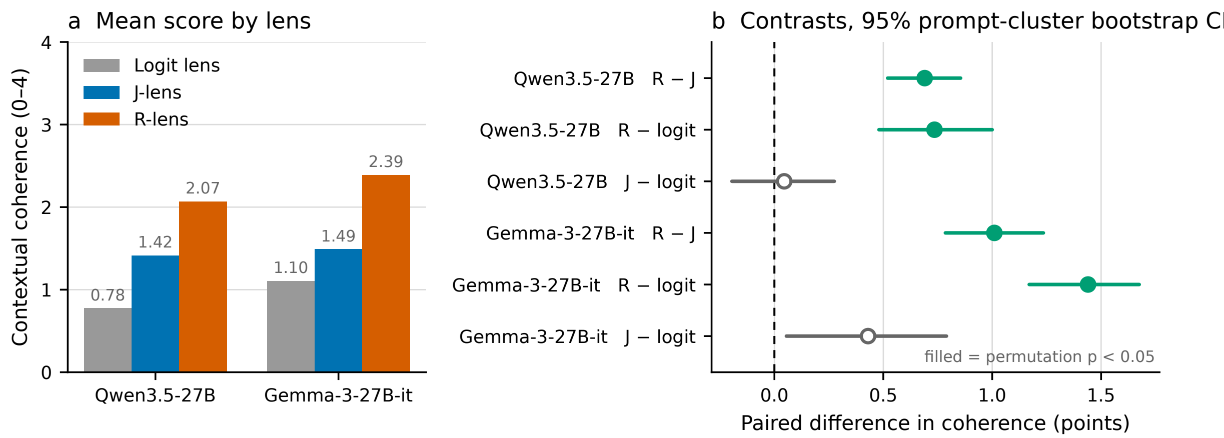

Figure 1: Primary result. (a) Mean contextual coherence by lens on the full 0–4 rubric scale. (b) Paired contrasts with 95% prompt-cluster bootstrap intervals; a filled marker denotes significance under the *pre-specified* sign-flip permutation test, which is not always the same as the interval excluding zero — J − logit on Gemma is drawn hollow for that reason. The axis in (a) is deliberately not rescaled: the ordering is clear, but every lens sits in the lower half of the scale.

Model	Contrast	Δ	95% CI	p	win	tie	loss
Qwen3.5-27B	R − J	0.690	[0.520, 0.855]	$< 10^{-4}$	.69	.20	.11
Qwen3.5-27B	R − logit	0.735	[0.480, 1.000]	$< 10^{-4}$	.64	.12	.24
Qwen3.5-27B	J − logit	0.045	[-0.195, 0.275]	0.776	.40	.19	.41
Gemma-3-27B-it	R − J	1.010	[0.785, 1.235]	$< 10^{-4}$	.76	.10	.14
Gemma-3-27B-it	R − logit	1.440	[1.170, 1.675]	$< 10^{-4}$	.80	.03	.17
Gemma-3-27B-it	J − logit	0.430	[0.055, 0.790]	0.112	.54	.06	.40

Table 1: Primary endpoint and the two supporting contrasts, under the mean-of-two-validated-judges rule. 100 cells and 20 prompts per model. Positive Δ favours the first arm.

the permutation test is the pre-specified one, so we read that contrast as not significant — but the disagreement is reported rather than settled by taking whichever of the two is more favourable.

Absolute scores are low. Mean contextual coherence is 1.99 (Qwen) and 2.57 (Gemma) for R-Lens, against 1.30 and 1.56 for J-Lens and 1.25 and 1.13 for the logit lens (Figure 1a). On this rubric the mid-scale band is "partially relevant, not a coherent continuation". R-Lens readouts at $z \le 0.4$ are better than the alternatives and still not good; on Qwen the R-Lens mean does not reach the midpoint of the scale.

4.2 Depth

The depth profiles differ between models. On Gemma the advantage holds at every depth and survives Holm correction at all five: +0.825 at $z = 0$, a peak of +1.325 at $z = 0.1$, +0.875 at $z = 0.4$. On Qwen it is largest early (+1.000 at both $z = 0$ and $z = 0.2$) and falls to +0.350 by $z = 0.4$; three of the five Qwen depths do not survive Holm correction ($z = 0.1$, 0.3 and 0.4, all p_{Holm} between 0.06 and 0.08).

The effect therefore decays with depth on Qwen and is roughly flat on Gemma, and on neither model is it concentrated at the single earliest layer. What both models support is that R-Lens is ahead across $z \le 0.4$; a sharper claim about *where* within early depth the improvement sits is not

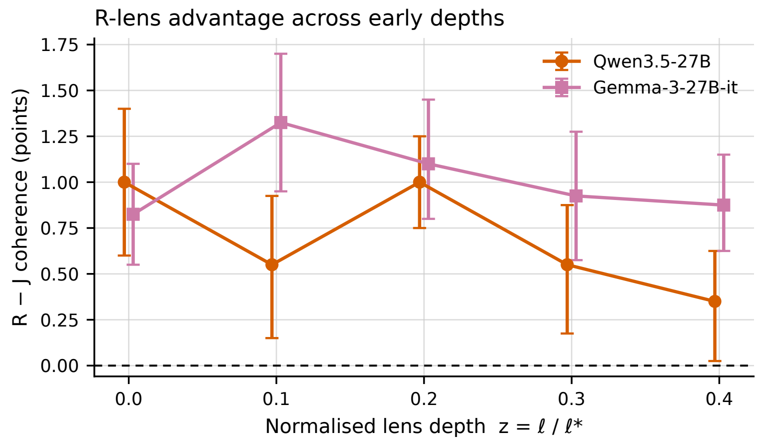

Figure 2: R − J by normalised depth, with 95% intervals. The two models do not agree on where in depth the advantage is largest.

supported by these two models together.

4.3 Secondary dimensions

Model	Dimension (R − J)	Δ	95% CI
Qwen3.5-27B	Lexical integrity	0.205	$[0.125, 0.275]$
Qwen3.5-27B	Prompt echo	0.245	$[0.160, 0.320]$
Gemma-3-27B-it	Lexical integrity	0.245	$[0.150, 0.345]$
Gemma-3-27B-it	Prompt echo	0.330	$[0.240, 0.410]$

Table 2: Secondary dimensions, both on 0–2 scales.

R-Lens produces more well-formed tokens (lexical integrity) *and* copies more of the prompt (prompt echo). Both differences exclude zero on both models. Section 5.4 asks whether the second explains the first.

We do not compare the magnitude of the echo difference to the magnitude of the coherence difference. They are measured on different scales (0–2 and 0–4) and such a comparison would be meaningless.

4.4 By evaluation set

Per-set differences are all positive on both models and range from +0.425 (Qwen typo; Gemma multilingual) to +2.150 (Gemma poetry). **No per-set difference survives Holm correction** ($p_{\mathrm{Holm}} = 0.641$ throughout). With four prompts per set this analysis is descriptive only and is reported for pattern, not for inference. Poetry is the largest difference on both models, but on four prompts that is an observation, not a finding.

5 Robustness

5.1 Integrity audit

A fail-closed audit re-derived the frozen artifacts from scratch: it recomputed the panel hash, replayed the per-(cell, rater) arm permutation and compared it to the key, reconstructed all 400 outgoing judge payloads and scanned them for lens-identity leakage, and re-parsed every stored judge response.

Twenty conditions were checked. On the first pass nineteen held—including zero unparseable responses across 526 stored ratings, zero arm-mapping mismatches, and an exact reproduction of the panel hash (`a4984e34d835c60ecb661552ec0a794e` …).

One condition failed, and it was ours. The score-combination rule derived the combined winner by *voting* on the judges' winners while *averaging* their scores, which let the two disagree on 18 of 200 cells. The rule was corrected to derive the winner from the combined scores, and the frozen raw responses were re-combined without any new API call, so no cell was re-rated and no rating changed.

Re-running the full analysis on the corrected scores changed no reported number—all 18 cells were ties under both rules. The audit was then re-run against the corrected ratings and the analysis those ratings produced, and **all twenty conditions pass**, including `winner_matches_max_score` across all 200 cells and `ratings_frozen_before_analysis`. A defect that changes no number is still reported here, because whether it changed a number is not something a reader should have to take on trust.

5.2 The adjudicator, and why it is not in the primary estimate

The original design adjudicated disagreements with a third judge. That judge was put through the same 78-cell battery as the primaries and cleared eight of nine gates outright: it never preferred a corrupted arm (0/10), showed no position bias ($\chi^2 = 0.66$ on 2 df), scored provably-identical arms identically, returned a tie on 5/5 identity-layer cells, showed full dynamic range on late-layer controls, and left 0/190 responses unparseable.

It did not clear the order-invariance gate, and—importantly—it did not fail it either. Rotating the arm order flipped the winner on **10 of 53 non-tied pairs (18.9%, 95% CI [9.4%, 32.0%])** against a 15% ceiling. An exact binomial test cannot reject 15% ($p = 0.266$), and the interval is far too wide to establish that the rate is acceptable. Escalating the battery does not help: resolving 18.9% against 15% at 90% power needs roughly 810 comparable pairs, a control panel several times the size of the experiment it exists to validate. For scale, the admitted primary judge GPT-5 scored 1/15 (6.7%) with an interval of $[0.2\%, 32\%]$—so this control cannot even separate the adjudicator from a judge already in use.

We therefore treated the unresolved gate as a failure and demoted the adjudicator, rather than admitting an instrument the battery could not clear or loosening the threshold to fit it. We do *not* claim to have shown the adjudicator is order-biased: the correct statement is that it did not satisfy a pre-specified validation gate, and the data cannot establish that its true rate is either acceptable or unacceptable. The primary estimate is the mean of the two validated judges. The adjudicated estimate is retained as a sensitivity analysis, and the adjudicator's ratings are preserved rather than deleted, since they are evidence about the panel even when not used to score it.

Disclosure of timing. The battery and its 15% ceiling were fixed in the frozen protocol before any judge was run. The *consequence* of excluding the adjudicator, however, was already visible:

the judge-dependence sensitivity analysis (§5.3), which reports the mean-of-two estimate at +0.850 against the adjudicated +0.778, was computed before the adjudicator's battery was run. We therefore describe the demotion as a transparent post-validation analysis amendment against a pre-specified gate, not as a fully blind fail-closed decision. Both estimates appear throughout.

5.3 Judge dependence

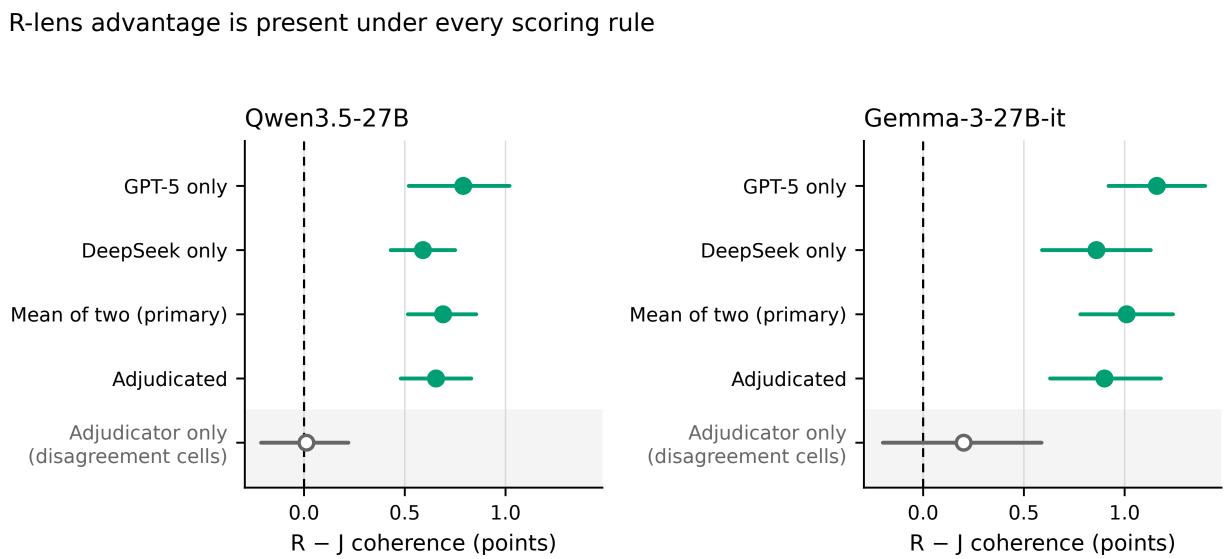

Figure 3: R − J under four scoring rules. "Mean of two" is the primary rule; "Adjudicated" is the original rule, retained as a sensitivity analysis after its third judge failed to clear validation. The shaded band is a diagnostic computed only on cells where the two primary judges disagreed; it is conditioned on disagreement and is not a fifth estimate of the same quantity.

A two-judge mean is still a scoring rule, and a different rule could give a different answer. Recomputing under four frozen rules—GPT-5 alone; DeepSeek alone; the mean of the two (primary); and the original adjudicated rule—gives a positive R − J on every rule and on both models, with every individual-judge interval excluding zero. The pooled estimate ranges from +0.725 (DeepSeek alone) to +0.975 (GPT-5 alone), with the primary rule at +0.850 and the adjudicated rule at +0.778.

The spread across rules (0.25 points pooled) is smaller than the effect itself, and no rule places the interval anywhere near zero. **The adjudicated estimate—the one built on the instrument we demoted—is the smallest of the four**, so excluding the adjudicator did not manufacture the result; it removed a conservative influence on it.

The logit-lens contrasts are not stable. That verdict is scoped to R − J, and the scoping is load-bearing. Checking every contrast against the same two single-judge intervals, five of nine are not stable across the two primary judges. On Qwen, GPT-5 rates J-Lens *significantly worse* than the logit lens (−0.680, [-1.06, -0.33]) while DeepSeek rates it significantly *better* (+0.770, [0.57, 0.97])—same cells, disjoint intervals, opposite signs. Every unstable contrast involves the logit lens; every R − J contrast is stable.

The natural reading is a floor effect: the judges agree about the relative ordering of two Jacobian transports and anchor the bottom of a 0–4 scale differently when asked to score near-noise. Any comparison against the logit lens in this report should be read with both judges' values in view and must not be quoted as a single number.

Agreement. Quadratic-weighted $\kappa = 0.514$ on 600 paired scores, exact agreement 39.3%, mean absolute difference 0.91 points. The two validated judges disagreed on 63% of cells—the disagreements the third judge was introduced to settle, and which the primary rule now simply averages over. This is moderate agreement at best. With validated judges it is evidence about the material rather than the instrument—the readouts being rated are genuinely ambiguous, which is what one expects of early-layer readouts—but a κ of 0.51 is not a precise instrument, and every interval in this document inherits that noise.

5.4 Prompt echo

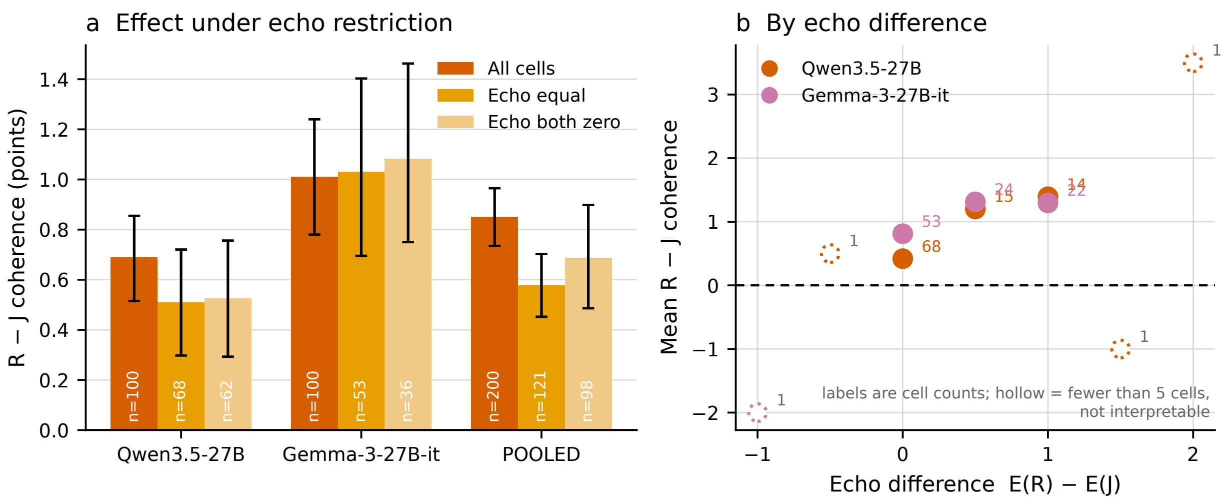

Figure 4: Echo sensitivity. (a) R − J restricted by echo agreement, retained cell counts printed inside each bar. (b) Mean R − J by echo difference; labels are cell counts and hollow markers mark strata too small to interpret.

R-Lens copies more of the prompt than J-Lens, and a rater who rewards prompt-shaped text would score it higher for that reason alone. Restricting to cells where both lenses received the *same* echo score:

Model	All cells	Echo equal	retained	Echo both zero	retained
Qwen3.5-27B	0.690	0.509 $[0.30, 0.72]$ (n=68)	74%	0.526 $[0.29, 0.76]$ (n=62)	76%
Gemma-3-27B-it	1.010	1.031 $[0.70, 1.40]$ (n=53)	102%	1.083 $[0.75, 1.46]$ (n=36)	107%
Pooled	0.850	0.577 $[0.45, 0.70]$ (n=121)	68%	0.686 $[0.49, 0.90]$ (n=98)	81%

Table 3: R − J contextual coherence under echo restriction, frozen mean-of-two scoring. "Retained" is the restricted estimate as a percentage of the all-cells estimate; it can exceed 100%, as it does on Gemma, where matching on echo does not shrink the effect at all.

The effect survives in all 24 subset estimates—every scoring variant, both models, both restriction rules—with intervals excluding zero throughout. But roughly a third of the pooled gap coincides with an echo difference under the equal-echo restriction, and the pattern is not uniform:

on Qwen the estimate falls by about a quarter, while on Gemma it does not fall at all. The Gemma both-zero cell retains only 36 cells from 17 prompts and its interval is correspondingly wide; it should not be read as evidence that echo is irrelevant there.

This is not a causal adjustment. Echo is measured from the same readouts as coherence, so restricting on it conditions on a post-treatment variable. The regression of the coherence difference on the echo difference is descriptive; its intercept is the fitted difference at equal echo, not an echo-adjusted effect. The two judges also disagree about the mechanism: GPT-5's slope is $\approx$ 1.0–1.1 with intervals excluding zero on both models, while DeepSeek's is 0.29 $[-0.16, 0.80]$ and 0.35 $[-0.03, 0.63]$, both covering zero. GPT-5 sees echo and coherence tightly coupled; DeepSeek sees almost no coupling. Both still rate R above J.

Read this analysis as a caution, not as reassurance. Restricting to cells with equal observed echo selects on a variable that lens identity itself affects. Conditioning on a post-treatment variable can distort a comparison and does not estimate the effect with echo removed. The following section measures the quantity directly, and gives a substantially less reassuring answer than this restriction does — which is the point of reporting both.

5.5 Non-echo coherence: direct measurement

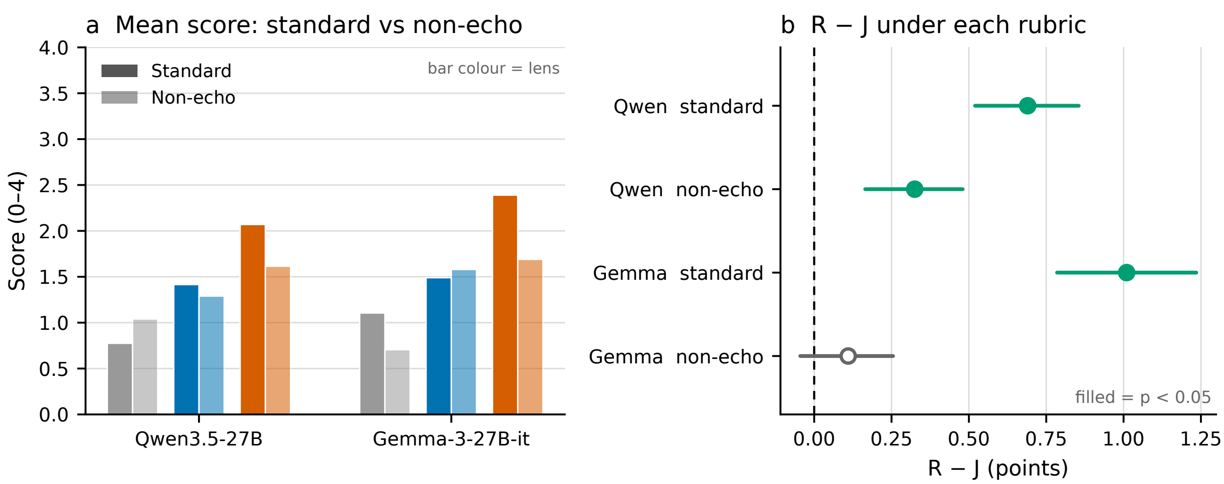

Figure 5: Standard versus non-echo scoring. (a) Mean score by lens under each rubric, same 0–4 scale. (b) R − J under each rubric; a filled marker denotes $p < 0.05$ on the pre-specified permutation test. The R-Lens bar falls sharply under non-echo scoring on both models while the J-Lens bar barely moves.

A second rubric was written, hashed and frozen under its own salt. It instructs the judge to discard every token that repeats a word visible in the prompt, including trivial variants, and to score only the remainder on the same 0–4 scale; a readout that is entirely repetition has nothing left and scores 0.

The rubric was validated before it was used. Ten control cells pitted a pure prompt-copy arm — the prompt's own final tokens — against real late-layer readouts. Both judges scored the copied arm **0.00 out of 4** and it won **0 of 10** cells. All 200 panel cells were then rated by both judges with zero parse failures. Rating was blocked in software until that control passed, and the control is keyed to the rubric's hash, so editing the rubric text invalidates its validation.

Model	Contrast	Δ	95% CI	p
Qwen3.5-27B	R − J	0.325	$[0.165, 0.480]$	0.0042
Qwen3.5-27B	R − logit	0.575	$[0.415, 0.715]$	$< 10^{-4}$
Qwen3.5-27B	J − logit	0.250	$[0.025, 0.465]$	0.070
Gemma-3-27B-it	R − J	0.110	$[-0.045, 0.255]$	0.450
Gemma-3-27B-it	R − logit	0.985	$[0.805, 1.180]$	$< 10^{-4}$
Gemma-3-27B-it	J − logit	0.875	$[0.685, 1.065]$	$< 10^{-4}$

Table 4: Non-echo coherence. Same estimand, estimator, bootstrap and permutation test as Table 1; a different scoring construct.

On Gemma the R-Lens advantage over J-Lens does not survive. The estimate falls from +1.010 to +0.110, the interval includes zero, and the paired win/tie/loss is 46/27/27 — a coin flip. On Qwen an advantage survives at less than half its size, $+0.690 \to +0.325$, still excluding zero.

We state these as *attenuations of the measured gap*, not as causal decompositions: the R − J gap attenuates by 53% on Qwen and 89% on Gemma under a rubric that excludes prompt-overlapping content. It does not follow that 53% or 89% of the original effect *is caused by* echo. The two rubrics measure different constructs, and the non-echo rubric additionally penalises a lens whose copied tokens occupy slots in a fixed-length top-10 list — an effect we could separate only by refilling the list from deeper rankings, which we did not do (§10).

What strengthens under non-echo scoring. Both Jacobian lenses beat the logit lens on both models, and J − logit is *larger* here than under standard scoring on Gemma (+0.875, $p < 10^{-4}$, against +0.430, $p = 0.112$). J-Lens echoes less than R-Lens, so standard scoring was penalising it for content the non-echo rubric ignores. The cross-model, cross-rubric result with the strongest support is therefore about the Jacobian transport in general, not about the LRP variant specifically.

The two echo analyses disagree, and the direct one should be believed. Restriction (§5.4) retained 74% of the effect on Qwen and 102% on Gemma. Direct measurement retains 47% and 11%. Conditioning on equal observed echo and excluding echoed content answer different questions: the former selects on a lens-affected variable, the latter changes the scoring construct. Neither identifies a formal mediation fraction.

5.6 Judge behaviour

R-Lens wins 69% (Qwen) and 76% (Gemma) of R − J cells and loses 11% and 14% respectively. The effect is a broad shift across the panel, not a handful of extreme cells — so it is not the product of a few outlier readouts that a different rubric might score differently.

6 Small-sample stability

The inferential unit is the prompt, not the cell: 20 prompts per model, each contributing five depths and three arms. One hundred cells can conceal a result that rests on one unusual prompt, so each prompt — and then each evaluation set — was deleted in turn and the estimate recomputed. Deleting a prompt deletes all of its cells, because the prompt is the bootstrap cluster.

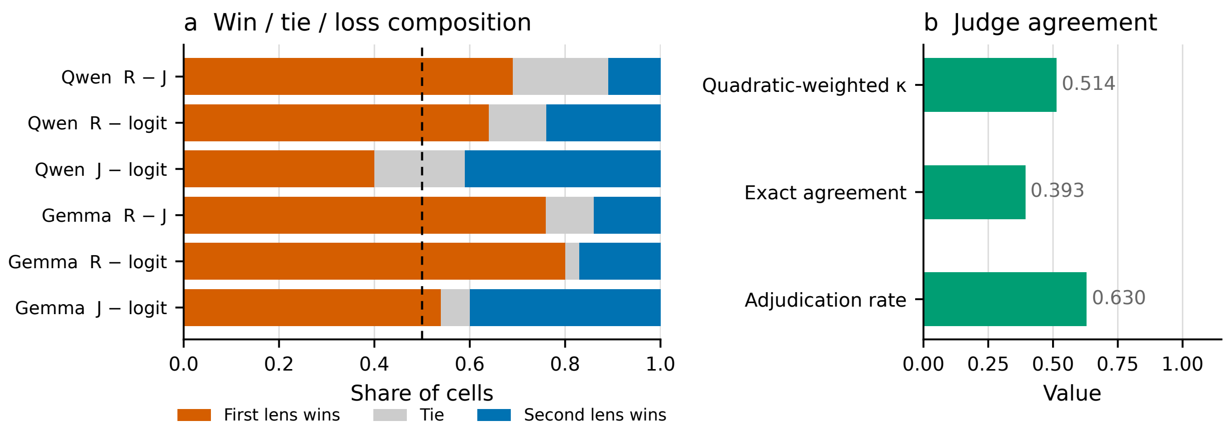

Figure 6: What the judges did. (a) Win/tie/loss composition per contrast. (b) Agreement statistics. A mean difference of 0.78 points is compatible with very different underlying distributions; the composition shows which.

Construct	Model	Leave-one-prompt-out	Leave-one-set-out	Prompt sign test
Standard	Qwen	0.600–0.700 (20/20 >0)	0.556–0.706 (5/5)	18+/2−, $p = 0.0004$
Standard	Gemma	0.820–0.973 (20/20 >0)	0.650–1.006 (5/5)	16+/2−, $p = 0.0013$
Non-echo	Qwen	0.273–0.380 (20/20 >0)	0.263–0.394 (5/5)	14+/5−, $p = 0.064$
Non-echo	Gemma	0.052–0.172 (20/20 >0)	−0.081–0.281 (4/5)	11+/5−, $p = 0.210$

Table 5: R − J under exhaustive single deletions. The sign test is a two-sided exact binomial over per-prompt effects, each averaged across that prompt's five depths; it is descriptive and discards magnitude, so it is less powerful than the primary estimator and is reported beside it, not instead.

The standard result is not driven by any one prompt or set. On both models R − J stays positive after deleting any single prompt and after deleting any single evaluation set, and the prompt-level sign test is significant on both. The n=20 objection does not undermine the standard finding.

The non-echo results are not equally stable, in two specific ways. First, poetry carries a disproportionate share of the Gemma effect. Removing it drops the standard Gemma estimate to +0.650, the lowest of the five deletions, and takes the non-echo Gemma estimate *negative* (−0.081). Poetry is also the set with the largest per-set difference on both models (Table in §4.4) and the only set read at a different position — the newline ending line one, per the source protocol, rather than the final prompt token. We do not have the prompts to determine whether that position, or the genre, is responsible.

Second, the prompt-level sign test does not confirm the Qwen non-echo result ($p = 0.064$, 14 of 19 non-tied prompts positive), even though the primary estimate excludes zero (+0.325, $[0.165, 0.480]$, $p = 0.0042$). The two are not in contradiction — the sign test discards magnitude and is the weaker instrument — but the Qwen non-echo advantage rests more on the size of the differences than on a clear majority of prompts, and should be described that way.

Taken together: the standard-rubric ordering is robust, and the non-echo findings are directionally consistent but fragile. That fragility is itself informative, since the non-echo construct is the one on which the headline claim was already weakest.

7 The earlier no-prompt panel

An earlier 50-cell panel on Qwen returned a null for R − J (+0.080, 95% CI $[-0.16, 0.34]$, $p = 0.596$). It is reported here for completeness and is **not comparable** to the present result. The two panels differ simultaneously in prompt visibility (that panel did not show raters the prompt), size (50 vs 200 cells), rubric scale, judge count, sampling rule and adjudication, and one of its raters was contaminated. No single one of those differences can be identified as the reason the results differ, and we do not claim to know which mattered. It is a preliminary attempt that was superseded, not a contradicting measurement.

That panel did produce one durable negative result, obtained against 150 human judgements: of three cheap token-form proxies for coherence, only the "trash-rate" proxy correlated in the expected direction ($\rho = -0.640$). Zero-corpus-frequency rate ($\rho = +0.330$) and in-prompt rate ($\rho = -0.245$) were both *inverted*, retracting two earlier readings of ours. Token-form statistics are not a substitute for rating the readouts.

8 Limitations

1. **One judge was demoted, not merely unused.** The adjudicator cleared 8 of 9 validation gates but its order-invariance rate (18.9%, $[9.4\%, 32.0\%]$) could neither be admitted nor rejected by a feasible battery. We excluded it. A reader who would have admitted it should read the adjudicated row of Figure 3 as the headline instead; it is smaller (+0.778 pooled) and still excludes zero.
2. **Order invariance is weakly controlled for every judge.** The same battery gives GPT-5 an interval of $[0.2\%, 32\%]$. The control is too blunt at this size to certify any of the three judges tightly, and a larger one was judged not worth its cost.
3. **Released lenses use the 25-prompt light recipe**, not the 1000-prompt recipe of the source post. Effect sizes are not comparable to the authors'.
4. **Twenty prompts per model.** Adequate for the pooled estimand, which is why intervals are prompt-clustered; not adequate for the per-set breakdown, which is descriptive only.
5. **Autoraters, not humans.** No human panel was run. Judges were validated against controls, but LLM raters may share biases with each other and with the models being read.
6. **Echo is not causally controlled**, only restricted on.
7. **Logit-lens comparisons are judge-dependent** and should not be quoted as single numbers.
8. **Depth-localisation does not replicate across models.**
9. **Ordering, not quality.** Nothing here shows early-layer R-Lens readouts are coherent in absolute terms; they average roughly 2 of 4, and under non-echo scoring less than that.
10. **The non-echo rubric penalises crowd-out as well as copying.** Copied tokens both earn no credit *and* occupy slots in a fixed-length top-10 list that novel tokens could have filled. Separating those would require refilling each list from deeper rankings; the frozen panel stored only the top 10, so we did not. The attenuations are therefore upper bounds on the loss of novel content, not clean estimates of it.
11. **Four prompts per evaluation set.** Per-set results are descriptive and none survives Holm correction.
12. **One rubric, one panel per construct.** We did not test whether a differently-worded non-echo rubric gives the same attenuation.
13. **Poetry is influential on Gemma.** It is the largest per-set difference on both models and the

only set read at a different position. Removing it takes the Gemma non-echo estimate negative. Four prompts is too few to say whether the position, the genre, or chance is responsible.

9 Supported and unsupported claims

Supported by these data.

1. Under the standard contextual rubric, R-Lens receives higher coherence scores than J-Lens on both models, and the direction holds for each validated judge separately.
2. R-Lens surfaces significantly more prompt-overlapping content than J-Lens on both models.
3. The R − J gap is substantially smaller under a rubric that excludes prompt-overlapping content.
4. On Qwen a smaller non-echo advantage survives; on Gemma the non-echo estimate is inconclusive.
5. Both Jacobian-based lenses outperform the logit lens on non-copied content, on both models. This is the most robust cross-model result here.
6. The standard-rubric R − J difference survives deletion of any single prompt and any single evaluation set, on both models (§6).

NOT supported, and not claimed.

1. That R-Lens is universally more semantically informative than J-Lens.
2. That a specific percentage of the standard R − J effect *is caused by* prompt echo. The attenuations are construct changes, not mediation fractions.
3. That restricting to equal observed echo identifies the direct effect of lens type. It conditions on a lens-affected variable.
4. That the source post's n=1000 fitting recipe was replicated. We evaluate the released n=25 light-recipe artifacts.
5. That autorater scores equal human judgement. No human rated anything here.
6. That J-Lens consistently beats the logit lens under the standard rubric. That contrast is judge-dependent and reverses in sign between judges.
7. That the excluded adjudicator was shown to be order-biased.
8. That a non-significant result establishes equality.
9. That five depths per prompt constitute independent observations. The inferential unit is the prompt: 20 per model.
10. That the non-echo findings are as stable as the standard ones. On Gemma the non-echo estimate turns negative when the poetry set is removed, and neither model's non-echo effect is confirmed by a prompt-level sign test at $\alpha = 0.05$.

10 What would change the conclusion

- The adjudicator failing its validation battery would not overturn the direction, but would demote the adjudicated estimate and promote the mean-of-two.
- A refilled non-echo analysis — filtering prompt copies from a deeper ranking and topping the list back up to ten — would separate loss of novel content from top-10 crowd-out. If the Gemma null persisted after refilling, the reading would firm up to "R-Lens has no detectable advantage in novel content on Gemma". If it became positive, the attenuation would be largely a presentation effect.

- A human panel disagreeing with the autoraters at scale would invalidate the instrument, not merely widen the intervals.
- Refitting both lenses at $n = 1000$ and finding the gap disappears would indicate the effect is a property of the light recipe.

11 Reproducibility

Every number in this document is produced by a command in the repository operating on frozen artifacts; none is transcribed by hand into a plotting script or a table. The figures are drawn by `rlens figures` directly from the analysis outputs, and skip themselves by name if an input is missing rather than falling back to defaults.

Stage	Command
Preflight (fail-closed)	`rlens preflight`
Prompt selection	`rlens eligibility`
Panel freeze	`rlens freeze-panel`, `rlens panel-v2`
Judge validation	`rlens judge-validate`
Rating	`rlens autorate`
Primary analysis	`rlens analyse-v2`
Integrity audit	`rlens audit-v2`
Judge and echo sensitivity	`rlens robustness`
Figures	`rlens figures`

Artifact hashes, the protocol salt, seeds and the audit verdicts are recorded in `final_reproducibility_manifest.json`. The unblinding key is deliberately absent from the repository and from this document's source tree.